

Analysis of SAR Images of Wind Streaks Over the Ocean

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Introduction

Synthetic Aperture Radar (SAR) Imaging:

- ▶ SAR is an active data collection technique where an instrument sends out a pulse of energy and records the amount of that energy reflected back after interacting with Earth's surface.
- ▶ SAR imagery is created from the interaction of an emitted pulse of energy with physical structures and environmental conditions.
- ▶ Wind streaks on the ocean typically have wavelengths ranging from hundreds of meters to a few kilometers depending on the wind speed and fetch (the distance over which the wind blows); these can be observed through satellite imagery using SAR.
- ▶ A typical wavelength for an open ocean wave (not wind streaks) is around 100 meters.

Project Objective:

- ▶ Analysis of wind streaks over the ocean captured in SAR images.
- ▶ Focus on **calculating the high-frequency and low-frequency wavelengths** associated with these wind streaks.
- ▶ **Why is it Important?**
- ▶ Wind streaks observed in SAR images are indicators of wind-driven surface currents and wave dynamics.
- ▶ Calculating the wavelengths helps us understand the dominant wind patterns.
- ▶ These insights can contribute to better oceanographic modeling and climate studies.

Dataset Overview

- ▶ Sentinel-1 Mission: Part of the Copernicus initiative by the European Commission (EC) and European Space Agency (ESA).
- ▶ Consists of two sun-synchronous, polar-orbiting satellites providing SAR radar imagery for environmental monitoring.
- ▶ Wave Mode Imaging:
 - ▶ Captures data in 20 km x 20 km vignettes with a 5 m x 5 m spatial resolution.
 - ▶ [Dataset](#): Contains over 37,000 SAR vignettes focusing on oceanic and meteorological phenomena.
 - ▶ Data is available in GeoTIFF and PNG formats;
 - ▶ PNG images were used for this project due to easier access, though they offer lower resolution than GeoTIFFs.

GUI and Algorithm Overview

Steps in Preprocessing of Data

- ▶ Image Upload: User selects and uploads an image, which is displayed in the UI.
- ▶ Metadata Extraction: Image dimensions are extracted, and spatial resolution is calculated and displayed.
 - ▶ There is an option to insert image resolution, but the default is 20km x 20km.
- ▶ Grayscale Conversion: Converts RGB images to grayscale; grayscale images are used as-is.
- ▶ Dimension Adjustment: Ensures image dimensions are even by adjusting height and width if necessary (helps with easier manipulation, especially in Fourier Transform's DC component).
- ▶ Display Preprocessed Image: Shows the cropped image in the UI and updates the dimensions in the interface.

MATLAB App

Upload Image

Pre-Process

Fourier Transform

25 50 75

Auto Detect/ Save Peaks

Display Saved Frequencies

Clear Saved Frequencies

Image Res. X [km] 20 Image Res. Y [km] 20

Pixel Res. 35.15m X 40.82m

Wind Streaks Image

Show Filtered Image High Freque...

Width 569

Height 490

Find Peak Frequency

Peak Freq X [1/m] 0

Peak Freq Y [1/m] 0

Wavelength [m] 0

Direction [degrees] 0

Intensity 0

MATLAB App

Upload Image

Pre-Process

Fourier Transform

25 50 75

Auto Detect/ Save Peaks

Display Saved Frequencies

Clear Saved Frequencies

Image Res. X [km] 20 Image Res. Y [km] 20

Pixel Res. 35.15m X 40.82m

Pre-Processed Image

Show Filtered Image High Freque... ▼

Width 568

Height 490

Find Peak Frequency

Peak Freq X [1/m] 0

Peak Freq Y [1/m] 0

Wavelength [m] 0

Direction [degrees] 0

Intensity 0

Fourier Transform Process

- ▶ Fourier Transform: Perform 2D Fast Fourier Transform (FFT) on the cropped grayscale image to transform it from the spatial domain to the frequency domain.

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

- ▶ This transforms the image from the spatial domain $f(x, y)$ to the frequency domain $F(u, v)$.
- ▶ **Frequency Spectrum Shift:** Shift the zero-frequency component to the center of the spectrum using `fftshift`.

$$\text{DC Component Location} = \left(\frac{M}{2}, \frac{N}{2} \right)$$

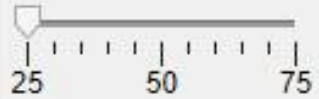
- ▶ **Magnitude Spectrum Visualization:** Compute the magnitude spectrum and apply a logarithmic transformation for visualization:

$$\text{Magnitude Spectrum} = c \cdot \log(|\text{Shifted Spectrum}| + 1)$$

Upload Image

Pre-Process

Fourier Transform



Auto Detect/ Save Peaks

Display Saved Frequencies

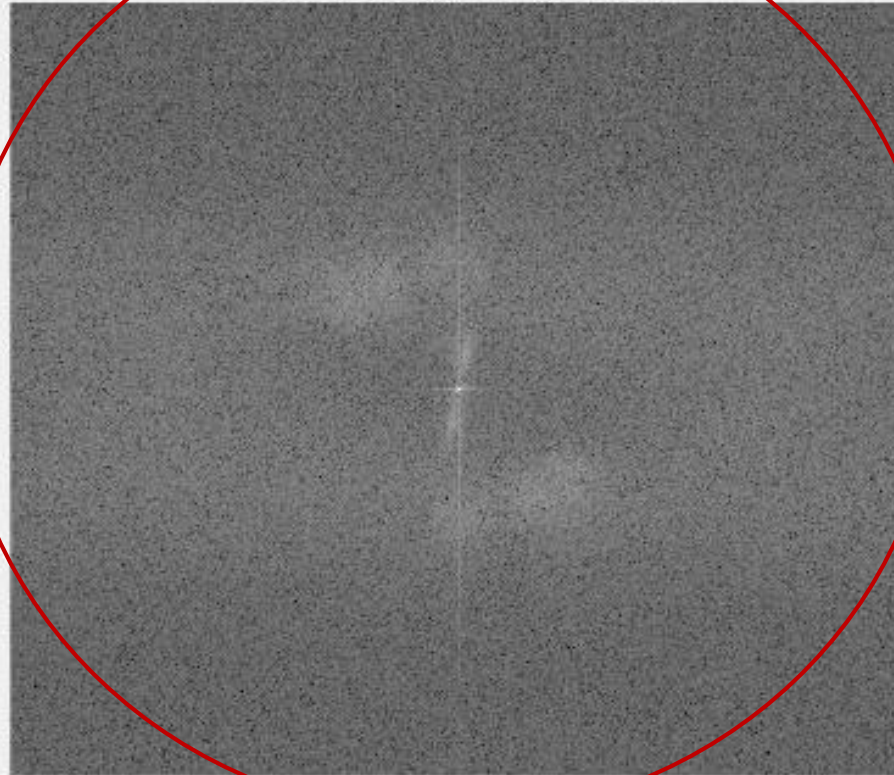
Clear Saved Frequencies

Image Res. X [km] 20

Image Res. Y [km] 20

Pixel Res. 35.15m X 40.82m

Fourier Transform of Image



Show Filtered Image

High Freque... ▼

Width 568

Height 490

Find Peak Frequency

Peak Freq X [1/m] 0

Peak Freq Y [1/m] 0

Wavelength [m] 0

Direction [degrees] 0

Intensity 0

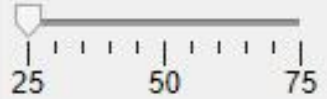
Manual Peak Frequency Search Algorithm

- ▶ User Range Selection: The user manually selects a range on the Fourier-transformed image.
- ▶ Search within Selected Range: The algorithm extracts the magnitude spectrum within the user-selected range.
- ▶ Find Maximum Peak: The algorithm searches within the cropped region and identifies the maximum peak (dominant frequency).
- ▶ Frequency and Direction: The coordinates of the peak are converted into frequency components (f_x , f_y).
- ▶ Wavelength and direction are calculated from these components.
- ▶ The peak is displayed visually on the magnitude spectrum and can be saved for later use.
- ▶ Calculations are displayed on the right sidebar (frequency, wavelength, direction, intensity, etc.)

Upload Image

Pre-Process

Fourier Transform



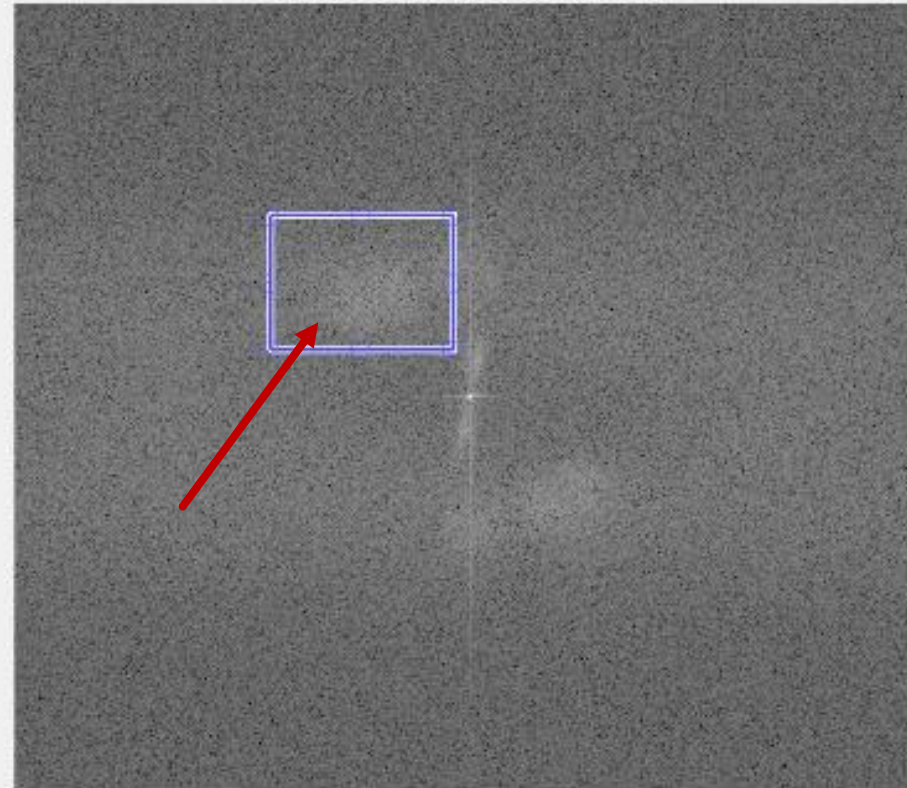
Auto Detect/ Save Peaks

Display Saved Frequencies

Clear Saved Frequencies

Image Res. X [km] Image Res. Y [km] Pixel Res.

Fourier Transform of Image



Show Filtered Image

High Freque... ▼

Width Height

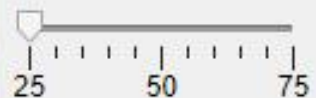
Find Peak Frequency

Peak Freq X [1/m] Peak Freq Y [1/m] Wavelength [m] Direction [degrees] Intensity

Upload Image

Pre-Process

Fourier Transform



Auto Detect/ Save Peaks

Display Saved Frequencies

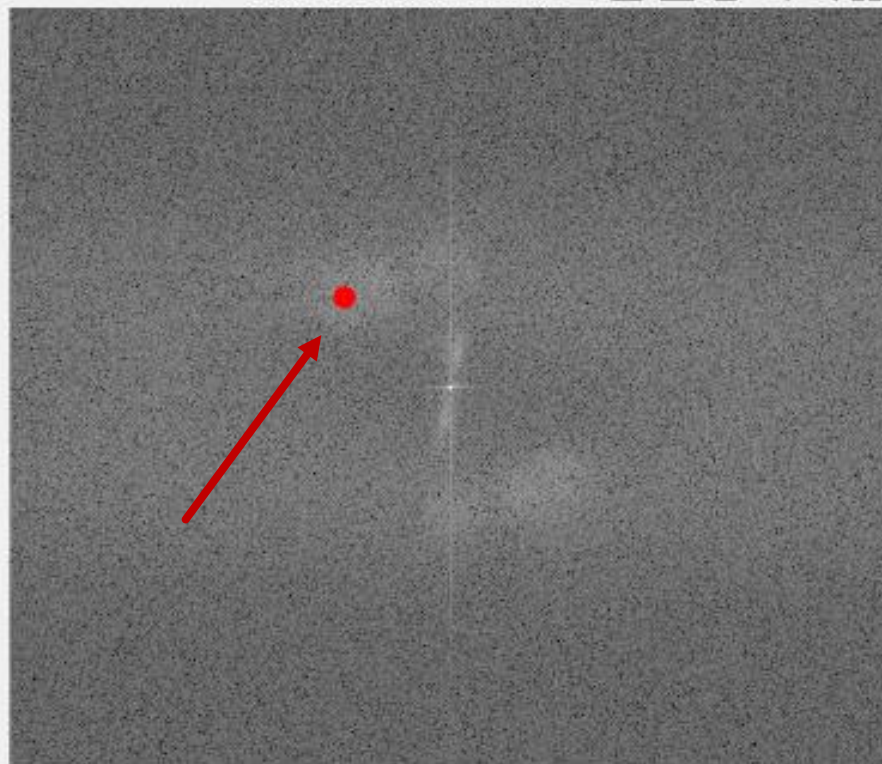
Clear Saved Frequencies

Image Res. X [km] 20

Image Res. Y [km] 20

Pixel Res. 35.15m X 40.82m

Fourier Transform of



Width 568

Height 490

Find Peak Frequency

Peak Freq X [1/m] -0.003356

Peak Freq Y [1/m] -0.00295

Wavelength [m] 223.8

Direction [degrees] -138.7

Intensity 62.53

☐ Save Frequency?

Automatic Peak Frequency Search Algorithm

- ▶ A low-pass (LP) Gaussian filter in the frequency domain is created:

$$H_{LP}(u, v) = \exp \left(-\frac{(u - \frac{N}{2})^2 + (v - \frac{M}{2})^2}{2\sigma^2} \right)$$

where u and v are frequency indices, N and M are image dimensions, and σ controls the filter width.

- ▶ The high-pass (HP) image is obtained by subtracting the LP filter from the original image:

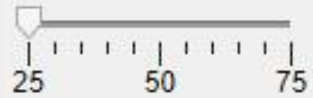
$$H_{HP}(u, v) = 1 - H_{LP}(u, v)$$

- ▶ The image is passed through both filters to create a high frequency image and a low frequency image.
- ▶ A **very** low pass filter is applied to the low frequency image to avoid leakage from the DC component.
- ▶ Maximum values are found in each window, and their average is computed for both LP and HP images.
- ▶ The peaks are displayed visually on the magnitude spectrum and are saved for later use. Their calculations are printed in the command window of Matlab.

Upload Image

Pre-Process

Fourier Transform



Auto Detect/ Save Peaks

Display Saved Frequencies

Clear Saved Frequencies

Image Res. X [km]

20

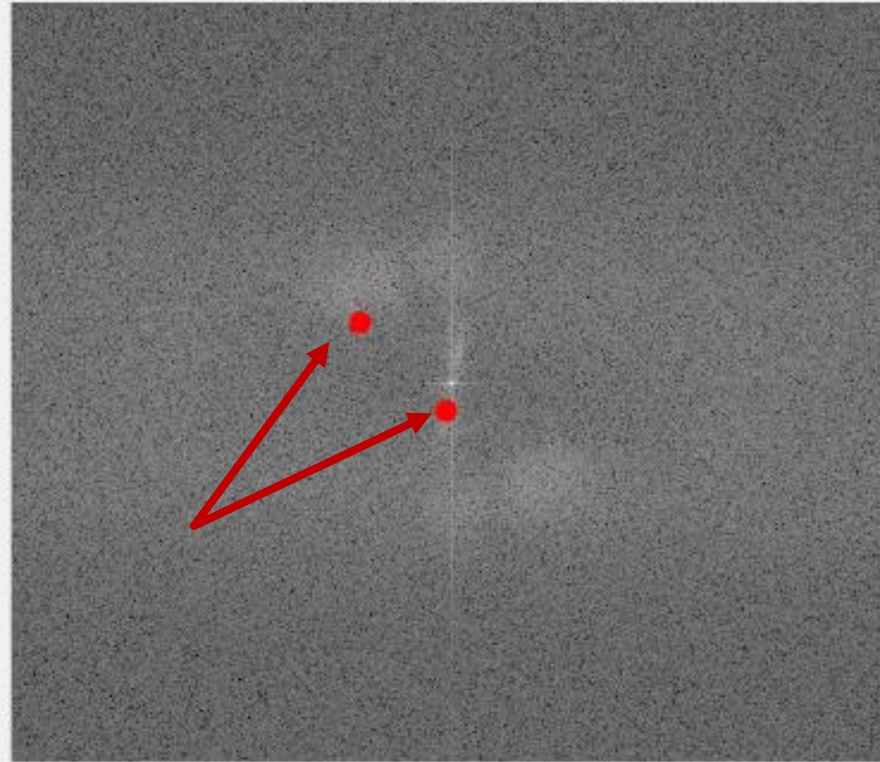
Image Res. Y [km]

20

Pixel Res.

35.15m X 40.82m

Fourier Transform of Image



Show Filtered Image

High Freque... ▼

High Frequency Average Peak:

Frequency X: -0.0019 Hz, Frequency Y: -0.0025 Hz

Wavelength: 312.4509 m

Direction: -127.5011 degrees

Intensity: 47.0727

Low Frequency Average Peak:

Frequency X: -0.0001 Hz, Frequency Y: 0.0009 Hz

Wavelength: 1066.1784 m

Direction: 96.9689 degrees

Intensity: 55.8063

Frequency and Wavelength Calculations

- ▶ X and Y frequencies are calculated as follows:

$$f_x = \frac{u}{M \cdot \Delta x} \quad f_y = \frac{v}{N \cdot \Delta y}$$

- ▶ u, v are the frequency indexes
- ▶ M, N is the total number of points (pixels).
- ▶ $\Delta x, \Delta y$ is the spatial resolution (distance between pixels).
- ▶ Since the fft was shifted, so DC is at the center,

$$u = coordinate_x - \frac{M}{2}, \quad v = coordinate_y - \frac{N}{2}$$

Frequency and Wavelength Calculations Cont.

- ▶ Wavelength and Direction are calculated as follows:

$$\lambda = \frac{1}{f} = \frac{1}{\sqrt{f_x^2 + f_y^2}}$$

Where:

- ▶ λ is the wavelength in the same units as the spatial dimensions of the image (e.g., meters if the image pixel spacing is in meters).
- ▶ f_x and f_y are the spatial frequencies in the x and y directions.

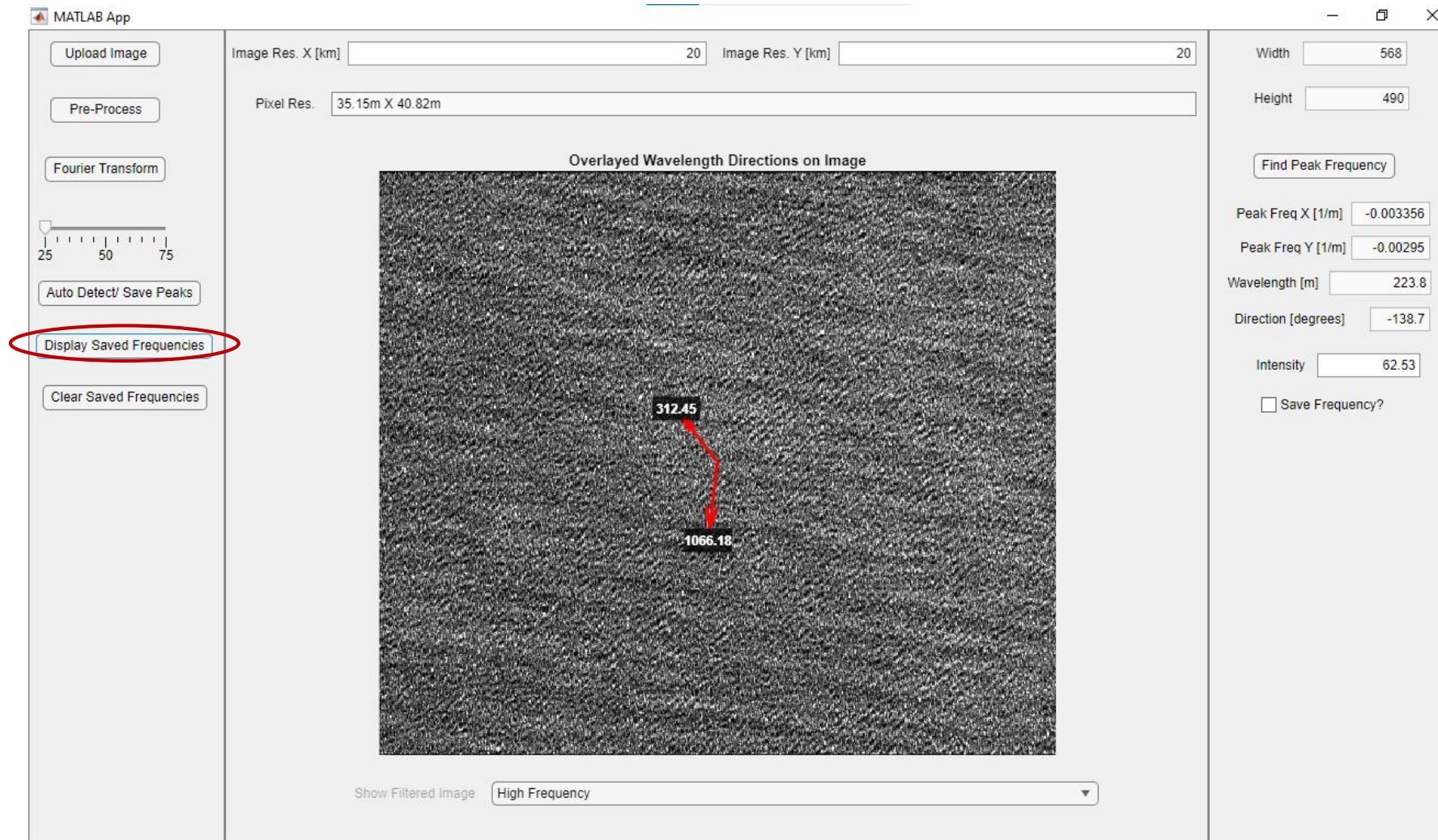
$$\theta = \tan^{-1} \left(\frac{f_y}{f_x} \right)$$

Where:

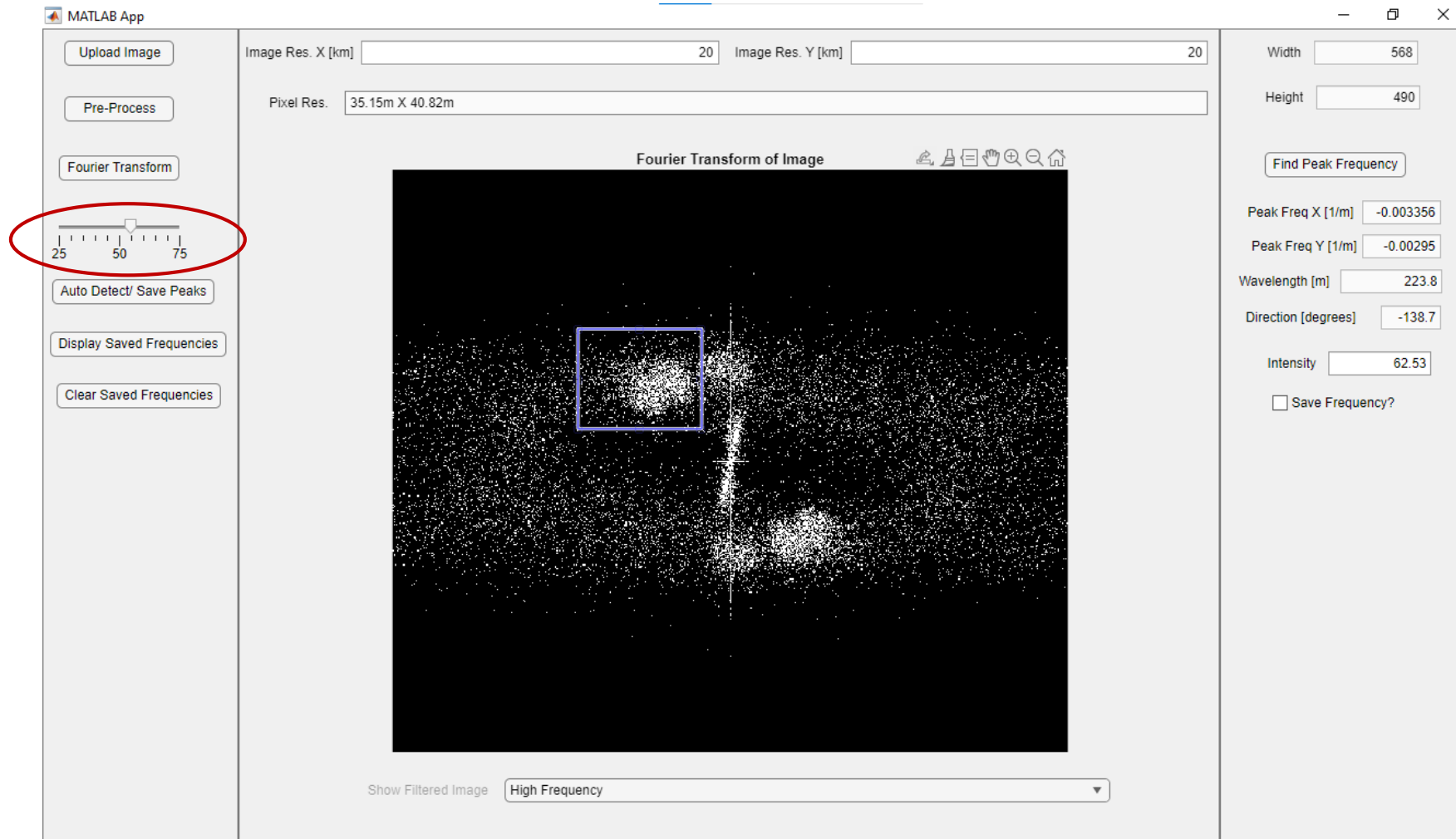
- ▶ θ is the direction of the wave relative to the x-axis, measured in radians or degrees.
- ▶ f_x and f_y are the spatial frequencies in the x and y directions.

Additional GUI Features

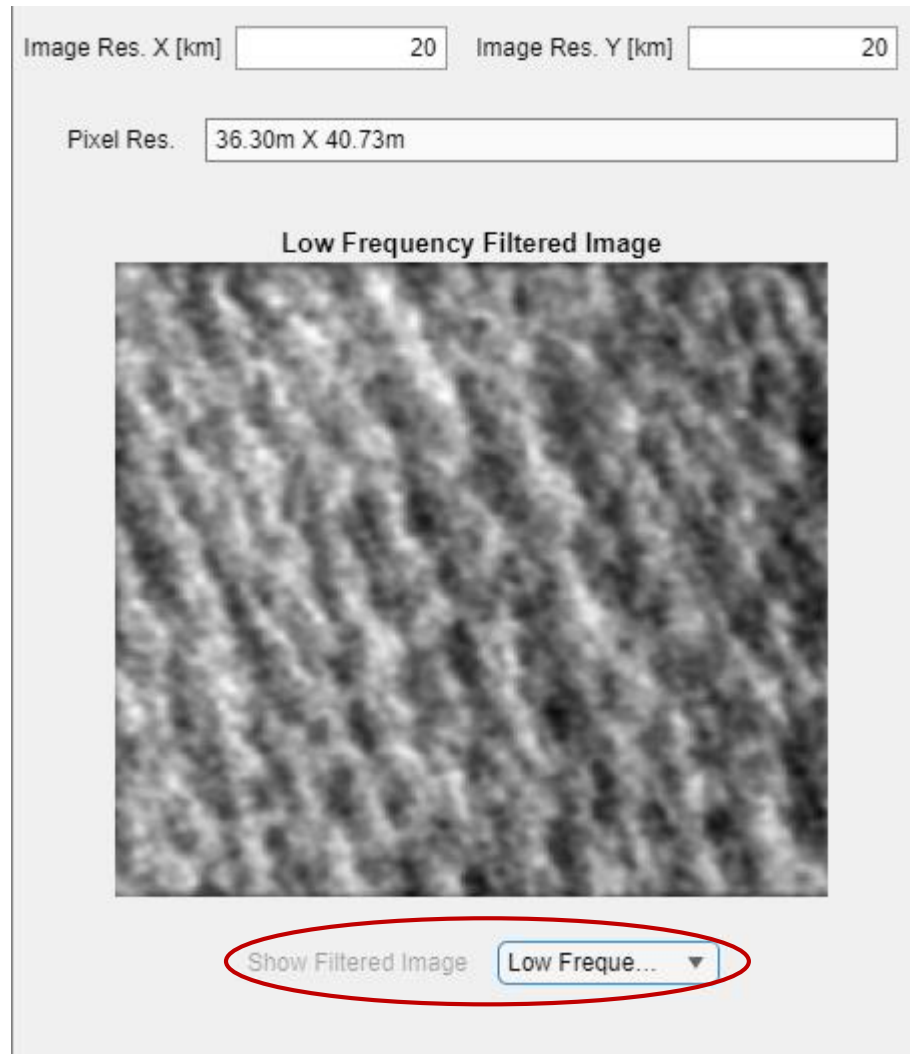
Display Saved Wavelengths and Directions Graphically



Use Binary Threshold on Fourier Transform for Ease in Finding Peaks

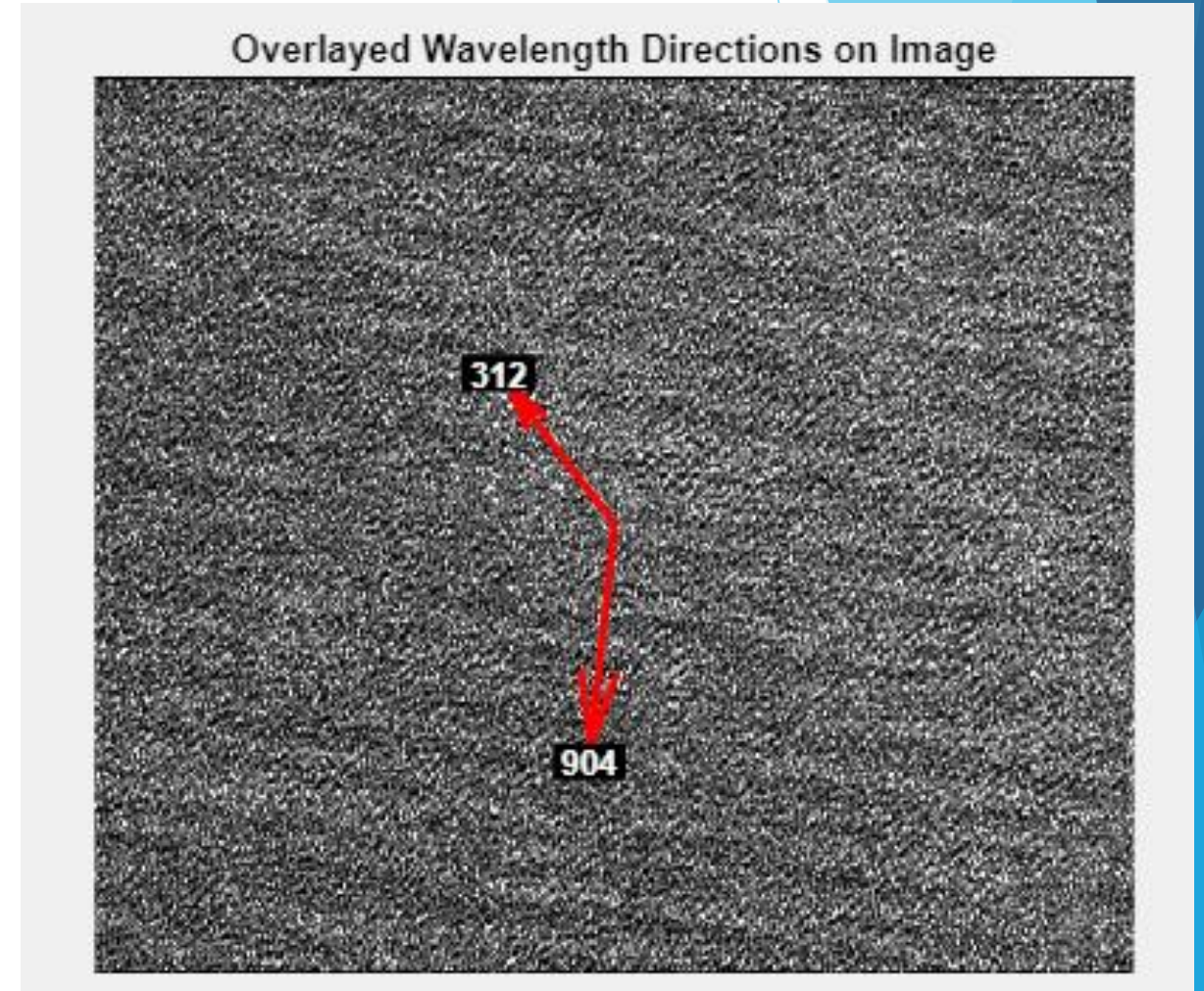
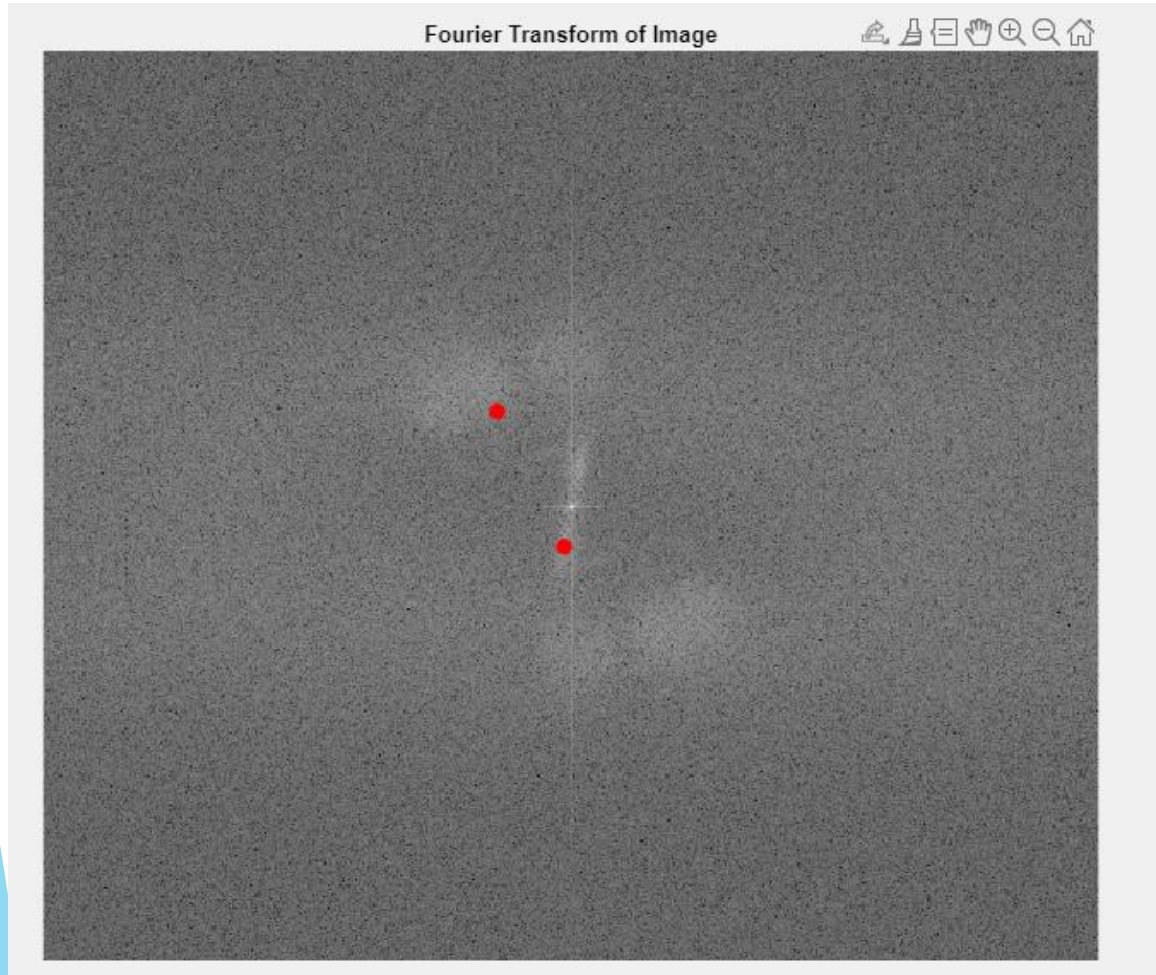


Display Filtered Low Frequency and High Frequency Images



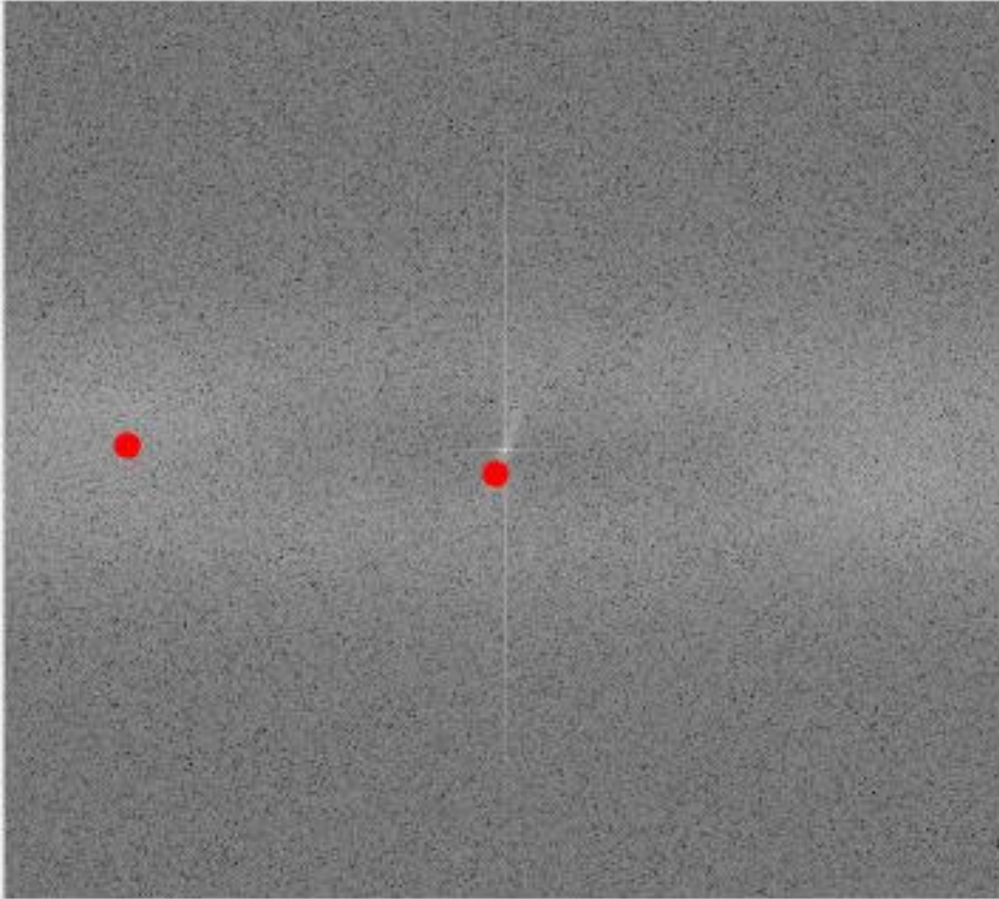
Results

Shorter Low Frequency Wavelength- Lower Wind Speed

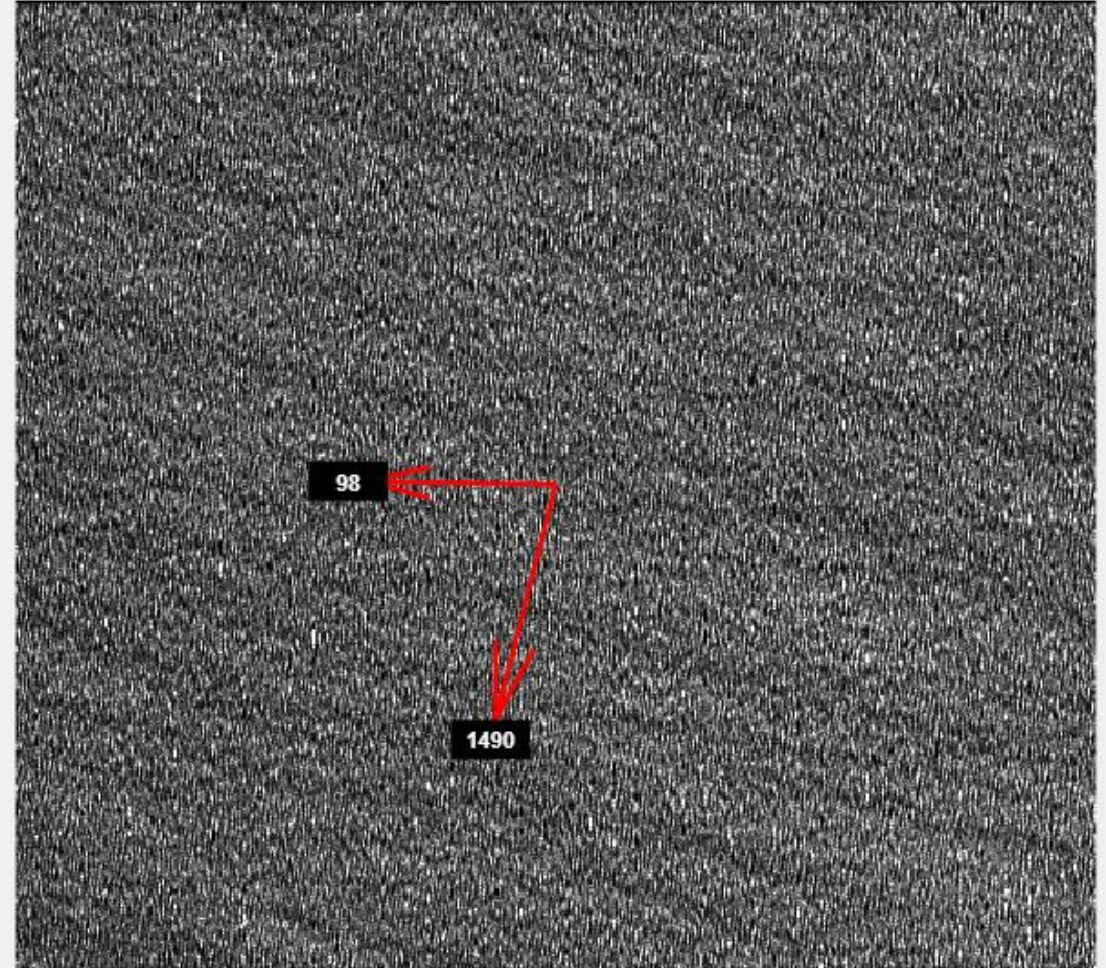


Mid Low Frequency Wavelength

Fourier Transform of Image

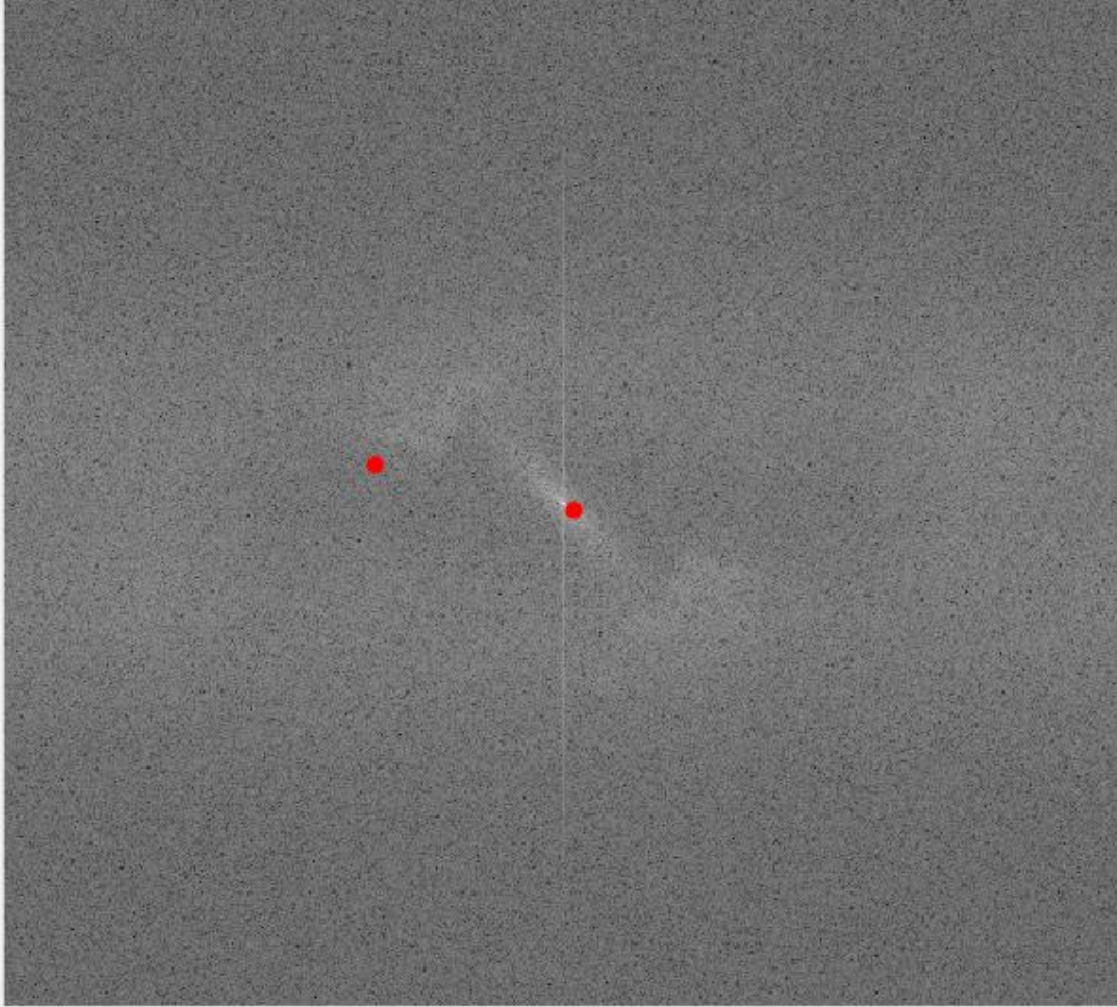


Overlaid Wavelength Directions on Image

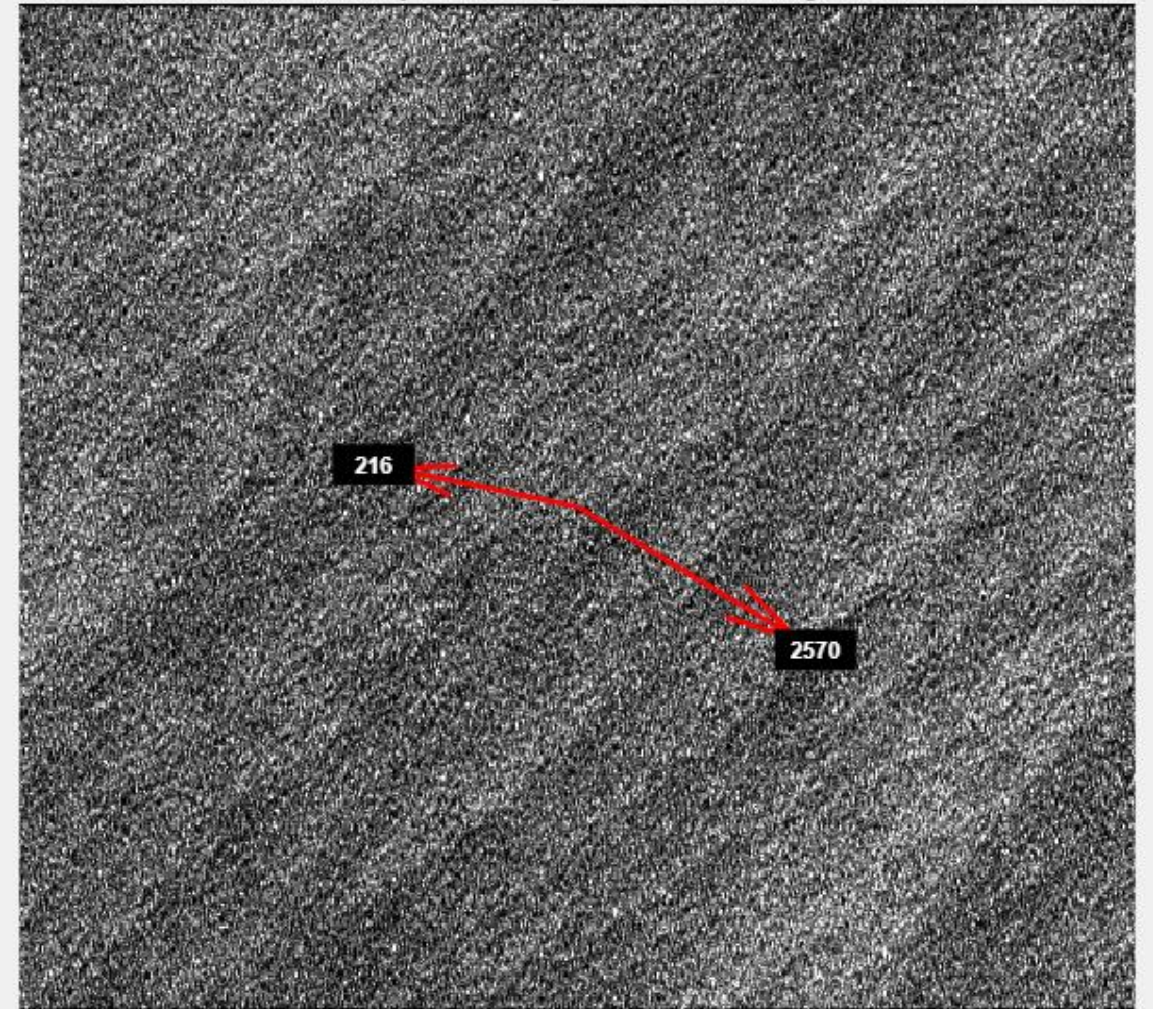


Longer Low Frequency Wavelength- Higher Wind Speed

Fourier Transform of Image



Overlaid Wavelength Directions on Image



Challenges Faced

- ▶ Lower Resolution in PNG Images:
 - ▶ Limited detail makes identifying precise frequency peaks difficult.
- ▶ Spatial Frequency Variability:
 - ▶ Frequency components vary across the image, creating a "cloud" of peaks.
 - ▶ Solution: Windowing technique was used to average peaks across different image regions for consistency.
- ▶ Low-Frequency Challenges:
 - ▶ Peaks close to the DC component are hard to distinguish from DC leakage.
 - ▶ Solution: A zeroing mask with a small radius was applied to suppress DC leakage, preventing it from being misidentified as a frequency peak.
 - ▶ Inverse relationship between frequency and wavelength: Small shifts in low-frequency values cause drastic changes in wavelength.
- ▶ Parameter Tuning:
 - ▶ Determining optimal window lengths and Gaussian filter width (sigma) required experimentation.
 - ▶ Solution: Windows were adjusted iteratively, and filtered images were checked to ensure effective isolation of high- and low-frequency components.
 - ▶ Still, small variations in parameters (such as window size, Gaussian filter width, and the radius of the zero mask around the DC component) had a significant impact on the identified peak frequency, leading to large variations in results.

Conclusions

- ▶ The automatic peak detection algorithm identified the magnitude and direction of peak frequencies for both the low-frequency and high-frequency components in the wind streak SAR images.
- ▶ **Low-frequency wavelengths (longer wind streaks):**
 - ▶ Ranged between 900 m to 2.6 km, aligning with theoretical predictions.
 - ▶ Precise magnitude identification was hindered by the low resolution of PNG images and the influence of the DC peak leakage.
- ▶ **High-frequency wavelengths (typical ocean waves):**
 - ▶ Ranged from 50 m to 400 m, as expected for ocean waves.
 - ▶ Similar wavelength patterns were observed in SAR images of open ocean without wind streaks, supporting the results.

References

- ▶ Dataset: <https://www.seanoe.org/data/00456/56796/>
- ▶ Synthetic Aperture Radar (SAR): <https://www.earthdata.nasa.gov/learn/earth-observation-data-basics/sar>
- ▶ Sentinel-1 mission: <https://sentiwiki.copernicus.eu/web/s1-mission#S1-Mission-Wave>
- ▶ Wind Streaks Wavelengths:
https://www.esa.int/esapub/eoq/eoq59/HORSTMANN.pdf?utm_source=chatgpt.com
- ▶ Open Ocean Waves Typical Wavelengths:
https://geo.libretexts.org/Bookshelves/Geology/Physical_Geology_%28Earle%29/17%3A_Shorelines/17.01%3A_Waves?utm_source=chatgpt.com